

Christopher Douglas Robert Wyatt

Curriculum Vitae

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Profile

I started my scientific career studying phylogenetics and molecular evolution, which got me hooked on understanding how the complexity of life evolved. With the arrival of improved DNA/RNA sequencing technologies, this opened up a world of high quality empirical data to better understand molecular regulation and evolution.

I have had the privilege to work with most of these next-generation sequencing technologies (RNA-Seq, MS-Proteomics, Hi-C) to address key questions in the field of evolutionary and developmental biology. I am proficient in Unix, Perl and R, and have working experience with Python/Matlab. I am also proficient in writing higher-level workflows using Nextflow, which allows the sharing of reproducible, concise and scalable pipelines. I have presented my work at many major meetings across Europe and am confident communicating my ideas and science.

I completed my doctoral training at the Centre for Genomic Regulation (CRG, Barcelona), where I worked on alternative splicing regulation in the lab of Manuel Irimia and additionally collaborated with the labs of Peter Holland (Oxford), Nacho Maeso (CABD, Sevilla) and Bill Keyes (IGBMC, Strasbourg), working in the fields of transcriptional regulation, reprogramming, senescence and splicing regulation, producing a number of high impact publications and gaining a broad knowledge of many bioscience fields.

Since my PhD, I worked at UCL (lab of Seirian Sumner), including using machine learning to discover genes required for caste identification across wasps, and at Seqera Labs (the authors of Nextflow) producing high quality informatics pipelines for startup clients. I am now back at UCL, developing reproducible, nf-core-standard Nextflow pipelines for evolution and ecology — including building **Eco-Flow**, a community working to close “the omics gap in ecology”.

Education & Research Career

Pipeline Developer

Seqera Labs, Barcelona

June 2021 – Sept 2021

Writing Nextflow pipelines for several startups as well as documenting the training material for Nextflow. This gave me in-depth knowledge of this domain specific language for pipeline development.

Post-doctorate in Evolutionary Genomics / Transcriptomics

Centre for Biodiversity and Environment Research, UCL, London

April 2019 – Feb 2021

Worked on a variety of research projects in the Eusocial Insect Research Group (Seirian Sumner) using machine learning, transcriptomics and proteomics approaches (using Nextflow) to explore wasp caste identity and evolution. Work submitted to bioRxiv prior to publication.

PhD in Biomedicine

Systems Biology, Centre for Genomic Regulation, Barcelona

Sept 2015 – March 2019

Studied the regulation and patterns of alternative splicing during pre-implantation mammalian development.

Selected Projects

Eco-Flow — open-source Nextflow pipelines for agri-ecology

eco-flow.github.io · github.com/eco-flow

UCL, ongoing

Contributing to a BBSRC-funded community building reproducible, nf-core-standard Nextflow pipelines that close “the omics gap in ecology”, alongside nf-test testing, researcher training, and community events.

Technical Skills

Languages & tools: Unix/Bash, Perl, R, Python, Matlab

Workflows: Nextflow (nf-core), reproducible & FAIR pipelines

Data types: RNA-Seq, MS-Proteomics, Hi-C, genome assembly

Publications

1. D. Sampaio Gonçalves, L. Dulac, B. Ritschka, M. Rhinn, A. Klein, **C.D.R. Wyatt**, M. Irimia & W.M. Keyes (2025). Senescent cells exhibit features of developmental signaling centres. *bioRxiv* (preprint). doi:10.1101/2025.10.30.685565
2. T.D. Doyle, E. Jimenez-Guri, J.C. Barnes, C. Hannah, S. Murray, **C.D.R. Wyatt**, O.M. Poole & K.R. Wotton (2025). Long-range pollen transport across the North Sea: Insights from migratory hoverflies landing on a remote oil rig. *Journal of Animal Ecology*, 94(11), 2267-2281. doi:10.1111/1365-2656.70126
3. A. Di Matteo, E. Belloni, D. Pradella, A.M. Chiaravalli, G.M. Pini, M. Bugatti, R. Alfieri, C. Barzan, E. Franganillo Tena, S. Bione, E. Terenzani, F. Sessa, **C.D.R. Wyatt** et al. (2023). Alternative Splicing Changes Promoted by NOVA2 Upregulation in Endothelial Cells and Relevance for Gastric Cancer. *International Journal of Molecular Sciences*, 24(9), 8102. doi:10.3390/ijms24098102
4. **C.D.R. Wyatt**, M.A. Bentley, D. Taylor, E. Favreau, R.E. Brock, B.A. Taylor, E. Bell, E. Leadbeater & S. Sumner (2023). Social complexity, life-history and lineage influence the molecular basis of castes in vespid wasps. *Nature Communications*, 14(1). doi:10.1038/s41467-023-36456-6
5. **C.D.R. Wyatt**, B. Pernaute, A. Gohr, M. Miret-Cuesta, L. Goyeneche, Q. Rovira, M.C. Salzer, E. Boke, O. Bogdanovic, S. Bonnal et al. (2022). A developmentally programmed splicing failure contributes to DNA damage response attenuation during mammalian zygotic genome activation. *Science Advances*, 8(15). doi:10.1126/sciadv.abn4935
6. **C.D.R. Wyatt**, M. Bentley, D. Taylor, R.E. Brock, B.A. Taylor, E. Bell, E. Leadbeater & S. Sumner (2020). Genetic toolkit for sociality predicts castes across the spectrum of social complexity in wasps. *bioRxiv* (preprint). doi:10.1101/2020.12.08.407056
7. E. Belloni, A. Di Matteo, D. Pradella, M. Vacca, **C.D.R. Wyatt**, R. Alfieri, A. Maffia, S. Sabbioneda & C. Ghigna (2019). Gene Expression Profiles Controlled by the Alternative Splicing Factor Nova2 in Endothelial Cells. *Cells*, 8(12), 1498. doi:10.3390/cells8121498
8. R.J. Southon, E.F. Bell, P. Graystock, **C.D.R. Wyatt**, A.N. Radford & S. Sumner (2019). High indirect fitness benefits for helpers across the nesting cycle in the tropical paper wasp *Polistes canadensis*. *Molecular Ecology*, 28(13), 3271-3284. doi:10.1111/mec.15137

9. F. Marlétaz, P.N. Firbas, I. Maeso, J.J. Tena, O. Bogdanovic, M. Perry, **C.D.R. Wyatt**, E. de la Calle-Mustienes, S. Bertrand, D. Burguera et al. (2018). Amphioxus functional genomics and the origins of vertebrate gene regulation. *Nature*, 564(7734), 64-70. doi:10.1038/s41586-018-0734-6
10. P. Kennedy, G. Baron, B. Qiu, D. Freitak, H. Helanterä, E.R. Hunt, F. Manfredini, T. O'Shea-Wheller, S. Patalano, C.D. Pull, T. Sasaki, D. Taylor, **C.D.R. Wyatt** et al. (2017). Deconstructing Superorganisms and Societies to Address Big Questions in Biology. *Trends in Ecology & Evolution*, 32(11), 861-872. doi:10.1016/j.tree.2017.08.004
11. I. Maeso, T.L. Dunwell, **C.D.R. Wyatt**, F. Marlétaz, B. Vető, J.A. Bernal, S. Quah, M. Irimia & P.W.H. Holland (2016). Evolutionary origin and functional divergence of totipotent cell homeobox genes in eutherian mammals. *BMC Biology*, 14(1). doi:10.1186/s12915-016-0267-0
12. M. Guillemin, L. Contreras-Porcia, M.E. Ramírez, E.C. Macaya, C.B. Contador, H. Woods, **C. Wyatt** & J. Brodie (2016). The bladed Bangiales (Rhodophyta) of the South Eastern Pacific: Molecular species delimitation reveals extensive diversity. *Molecular Phylogenetics and Evolution*, 94, 814-826. doi:10.1016/j.ympev.2015.09.027
13. S. Patalano, A. Vlasova, **C. Wyatt**, P. Ewels, F. Camara, P.G. Ferreira, C.L. Asher, T.P. Jurkowski, A. Segonds-Pichon, M. Bachman et al. (2015). Molecular signatures of plastic phenotypes in two eusocial insect species with simple societies. *Proceedings of the National Academy of Sciences*, 112(45), 13970-13975. doi:10.1073/pnas.1515937112
14. C. Giampietro, G. Deflorian, S. Gallo, A. Di Matteo, D. Pradella, S. Bonomi, E. Belloni, D. Nyqvist, V. Quaranta, S. Confalonieri, G. Bertalot, F. Orsenigo, F. Pisati, E. Ferrero, G. Biamonti, E. Fredrickx, C. Tavecchia, **C.D.R. Wyatt** et al. (2015). The alternative splicing factor Nova2 regulates vascular development and lumen formation. *Nature Communications*, 6(1). doi:10.1038/ncomms9479